

Gemini 3 赋能数学建模竞赛实战指南

从赛题解析到论文自动化

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当前主流大语言模型生态概览

在 2025 年的数模竞赛中，我们主要关注“第一梯队”的四个选手：

OpenAI (GPT-5 系列)

- **特点：**综合能力均衡，生态丰富。
- **局限：**上下文窗口较短 (128k)，处理大量附件时容易“遗忘”。

Google (Gemini 3)

- **特点：** **超大上下文 (2M Token)**，原生多模态，搜索结合紧密。
- **数模定位：**处理长文档与复杂数据的“重型武器”。

Anthropic (Claude Sonnet)

- **特点：**代码生成能力极强 (SOTA)，逻辑细腻。
- **局限：**每日使用次数限制严格，且多模态能力稍弱于 Gemini。

DeepSeek (R1/V3)

- **特点：**推理成本低，数学推导能力惊人 (O1 级别)。
- **定位：**适合纯数学推导，但多模态识图能力略逊。

为什么数模竞赛首选 Gemini 3? (能力对比)

针对数学建模的三大痛点，各模型表现如下：

能力维度	Gemini 3	GPT-5	Claude 3.5	DeepSeek
长上下文 (Reading) (一次读完 20 篇文献)	S+ (2M) 无遗忘	A (128k) 需截断	A (200k) 需截断	B (64k) 需截断
多模态 (Vision) (识别题目中的物理图)	S (原生) 视频/图	A+ 图片	A 图片	B- 仅文本
代码解释器 (Coding) (自修复运行代码)	A+ 内置沙箱	A+ 内置沙箱	S Artifacts	A 无沙箱

结论：虽然 Claude 写代码强，DeepSeek 算数学强，但 Gemini 3 是唯一的“全能六边形战士”，尤其在读题和处理附件上没有替代品。

Gemini 3 在数模中的“降维打击”能力

1. 无限的上下文窗口 (Infinite Context Window)

数模 C 题通常附带 5-10 个 CSV 文件和几百页的背景手册。

- **传统 LLM**：只能复制粘贴部分数据，容易丢失全貌。
- **Gemini 3**：可以直接把 **所有 PDF 和数据集** 扔进去，它能看到全局关联（例如：A 文件的第 5 列与 B 文件的第 3 列相关）。

2. 原生多模态思维 (Native Multimodality)

题目中常出现复杂的机械结构图或流体示意图。

- Gemini 3 不是“看图说话”，而是理解图像背后的**物理逻辑**，并直接将其转化为微分方程。

1.1 题目结构化拆解 (The Deconstructor)

场景：比赛刚开始，面对冗长的题目文本，需要快速提取核心要素。

Role: Expert in mathematical modeling.

Context: Uploaded problem description (PDF).

Task: Analyze and output in Markdown:

1. Background: Summarize in 3 sentences.
2. Key Requirements: List specific Questions (Q1, Q2...).
3. Variables: Create a Table [Symbol, Meaning, Unit, Type].

Prompt: Structure Extraction

教学要点

通过要求输出 **表格 (Table)** 和 **清单 (List)**，强制 AI 将非结构化文本转化为结构化数据，防止漏题。

1.2 文献与附件速读 (The Scanner)

场景：题目附带了多篇参考文献及多个数据附件，需要快速抽取可用信息。

```
Task: Read all attached reference documents.  
For each document, provide:  
- Core Concepts: Central ideas and theoretical basis.  
- Key Equations: Present in LaTeX form.  
- Methodology: Workflow and assumptions.  
- Relevance: How this document connects to the target problem.  
- Limitations: Concise evaluation of weaknesses.  
Ensure no hallucination and stay grounded in the text.
```

Prompt: Literature Understanding

```
Task: Analyze every attached dataset or supplementary file.  
For each file, extract:  
- Structure: Variables, dimensions, schema.  
- Data Types: Numerical, categorical, temporal, spatial, etc.  
- Distribution Insights: Highlight major patterns or anomalies.  
- Usage Potential: Which modeling tasks or visualizations it fits.  
- Constraints: Missing values, noise, sampling bias.  
Summaries must reflect the dataset exactly as provided.
```

Prompt: Data Attachment Understanding

教学要点

充分利用 Gemini 3 的 **Long Context** 能力，可同时解析数份文献与附件，形成可直接用于建模、复现实验或撰写 Related Work 的基础材料。

2.1 问题机理模型推导 (The Physicist)

目标：针对给定的具体问题 Q_x ，要求 LLM 从机理层面进行数学/物理建模，给出严谨的逻辑推导、变量定义与方程建立过程。

Task: Given the specific problem Q_x , construct a rigorous mechanistic model.

Requirements:

1. Interpret Q_x precisely and identify the underlying physical, mathematical, or dynamical principles that govern the system.
2. Define all variables, parameters, domains, and assumptions explicitly.
3. Derive the governing equations step-by-step with clear logic.
4. Provide strict reasoning, intermediate calculations, and justification for each transformation or approximation.
5. Output the final formulation in standard LaTeX (PDEs/ODEs/constraints).

Constraint: Do NOT solve the equations; only build and justify the model.

Prompt: Mechanistic Modeling & Logical Derivation

优势

使 LLM 不仅给出结果，还对问题 Q_x 进行深入结构化分析，形成逻辑自洽、计算清晰、推导严格的“模型建立”章节，可直接用于论文撰写。

2.2 数据清洗与可视化 (The Data Analyst)

场景：数据脏乱差（缺失值、异常值），需要快速预处理。

```
Context: Dataset `data.csv` with missing values.  
Task: Write & Execute Python code:  
1. Load data with Pandas.  
2. Fill missing values (linear interpolation).  
3. Detect outliers (Z-score > 3).  
4. Plot cleaned time-series.  
Constraint: Debug yourself if errors occur.
```

Prompt: Auto-Coding

关键指令：Debug Yourself

"Debug yourself" 是核心指令。它利用 Code Interpreter 的自愈能力，遇到报错自动修正，直到代码跑通。

2.3 智能求解 (The Optimizer)

场景：规划问题求解 (Linear Programming / Optimization)。

Task: Based on the model defined (Max Profit P).

Action:

1. Formulate **as** Linear Programming problem.
2. Use ``scipy.optimize`` **or** ``pulp`` to solve.
3. Print optimal decision variables.

Prompt: Optimization

2.4 绘图与示意图 (The Visualizer)

场景：论文中需要根据一段复杂的技术描述自动生成高质量的流程图或 Pipeline 示意图。

You are a senior scientific illustration specialist producing pipeline diagrams for top-tier journals. Use Banana-based textual reasoning to interpret the uploaded text. Extract workflow modules, data flow, dependencies, and hierarchy. Generate a clean, publication-grade pipeline figure.

Requirements:

- White background, no shadows/gradients.
 - Low-saturation scientific palette (CartoColor Muted/Antique, Tol Light/Muted).
 - Times New Roman, 11pt; italic numbers.
 - Bold connectors and arrows.
 - Diagram must strictly reflect the text's logic.
- Output: One high-resolution (600 dpi) schematic.

Prompt: Text-to-Pipeline Diagram

Task: Read the provided textual description and produce a publication-grade conceptual schematic.

Instructions:

- Identify components, interactions, and ordering.
 - Convert them into modular blocks + directional flow.
 - Use clean, academic design with low-chroma colors.
 - Maintain proportional spacing + clear labeling.
 - No invented steps; all elements must appear in the text.
- Output: A high-DPI workflow or system schematic figure.

Prompt: Conceptual Schematic from Text

教学要点

- **文本结构抽取：**依托 Banana 能力，Gemini 能从任意技术段落中自动解析流程结构并生成 pipeline 示意图。
- **顶刊美学规范：**使用低饱和科研配色、Times New Roman 字体、粗线条连接与高 DPI 输出，可直接达成 Nature/Science 水平图示质量。

2.5 顶级数据图直出 (The Direct Data Visualizer)

场景：不写代码，让 Gemini 读取数据文件，直接生成用于论文的专业统计图。

You are a senior scientific data visualization specialist producing publication-grade figures for Nature/Science. Based strictly on the real quantitative structure of the uploaded dataset, generate multiple high-resolution scientific figures (minimum 3), each using a different visualization type that best reveals a distinct aspect of the data.

Visualization rules:

- Background must be clean white, with no shadows, gradients, or decorative effects.
- Use a low-saturation, high-level scientific colormap (softened viridis/cividis/batlow).
- All text uses Times New Roman, 11pt; all numeric values must be italic.
- Axes must be boldened with heavier line weight.
- All labels, axes, and legends must be clear and consistent.
- Visual encodings (color intensity, line weight, marker size, density, etc.) must strictly correspond to the actual numeric magnitudes.
- You must not invent or infer values beyond the dataset.

Output:

Produce multiple distinct figures (3), each 600 dpi, each representing a different visualization category appropriate to the dataset. Include a short descriptive title for each figure based on what aspect of the data it represents.

Prompt: Direct Data-to-Image Generation

3.1 学术化润色 (The Polisher)

场景：将“中式英语”或口语化表达转化为 Top-tier 期刊风格。

Role: Academic Editor.

Input: "[Paste User Draft]"

Task: Rewrite for academic flow.

- Use passive voice for objectivity.
- Use precise vocabulary ("use" -> "utilize").
- Keep mathematical meaning unchanged.

Output: LaTeX text format.

Prompt: Tone Style Transfer

3.2 摘要生成 (The Summary Writer)

场景：截稿前 1 小时，快速生成结构清晰的 Summary Sheet。

Context: Problem=[X], Methods=[A, B], Results=[R].

Task: Write a 200-word Abstract (MCM format):

1. Para 1: Problem Restatement.
2. Para 2: Model Description & Justification.
3. Para 3: Validation & Sensitivity.
4. Para 4: Conclusion.

Tone: Professional and concise.

Prompt: Abstract Generation

实战目标：2025 MCM Problem C (奥运奖牌榜预测)

题目核心任务：

- **预测 (Prediction):** 建立模型预测 2028 洛杉矶奥运会奖牌榜。
- **量化 (Quantification):** 估算“伟大教练效应” (Great Coach Effect)。
- **分析 (Analysis):** 探讨“东道主优势”及新项目对奖牌的影响。

数据文件： summerOly_medal_counts.csv, hosts.csv, athletes.csv 等 5 个文件。

Gemini 3 破题策略

1. **秒读:** 提取 5 个 CSV 的关联字段。
2. **建模:** 自动选择时间序列模型 (ARIMA/LSTM)。
3. **制图:** 直出 2028 预测对比图。

总结：Gemini 3 速通核心流 (Summary)

- ① **Deconstruct:** 30 秒内从 PDF 提取出隐含约束（如不确定性分析）。
- ② **Auto-Code:** 一句话解决多表 Merge 和 IOC 代码清洗痛点。
- ③ **Reasoning:** 辅助构建“双重差分模型”来量化教练效应，不仅是做预测，更是做因果推断。
- ④ **Direct Vision:** 不写绘图代码，直接生成符合出版级的预测对比图。

Next Step: 动手尝试 2025 C 题!

Thanks!
提问环节